

Partnership Opportunity Request for Information (RFI)

Defense Microelectronics Activity (DMEA)

10 March 2025

1.0 SCOPE:

1.1 TITLE: Partnership Opportunity for Collaborative Manufacturing of Thick Silicon Carbide Epitaxial Layers for High Voltage DoD Applications

1.2 APPLICABILITY: Semiconductor Manufacturing

1.3 PROBLEM STATEMENT: The superior properties of Silicon Carbide (SiC) make the material attractive for use in both current and emergent DoD applications. SiC devices have the intrinsic capability to operate at higher temperatures, higher voltages, higher power densities, and higher frequencies than their silicon-based counterparts. These characteristics provide substantive benefit in DoD applications, including reduction in Size, Weight, and Power (SWaP), and enhanced survivability and reliability in harsh environments.

To maximize the benefit of SiC technology in DoD applications, it is necessary to produce thick epitaxial layers ($>100\mu\text{m}$) on 4H-SiC substrates, as this is a prerequisite for the fabrication of high voltage ($>10\text{kV}$) discrete components. These devices are leveraged in a host of DoD applications that require compact, high efficiency power conversion including medium-voltage converters, pulsed-power systems, and microgrids.

Very thick, low defect 4H-SiC epi layers on 6-in. wafers is a relatively low Technology Readiness Level (TRL) technology in the commercial sector. This is partially attributable to the lack of demand for ultra-high voltage components. Further, limited sources, low quality, and modest growth rates associated with current production thick epi SiC wafers contribute to exorbitant cost and are additional barriers that must eventually be overcome by the commercial sector to implement this technology. However, DoD demand for ultra-high voltage components requires immediate installation of thick-growth epitaxy capability to produce devices at a scale sufficient to meet DoD mission requirements.

2.0 DESIRED SOLUTION: DMEA seeks to establish an arrangement, such as: a Cooperative Research and Development Agreement (CRADA), a Commercial Service agreement, a Public-Private Partnership, or Other Transaction, as appropriate, with a qualified commercial entity that can demonstrate advanced technical knowledge and practical experience in SiC epitaxy. The primary objective of the proposed partnership is to establish a secure source of supply for high quality, thick epitaxial SiC on 150- or 200mm wafers that can be utilized by DoD programs. The potential commercialization of the technology developed under the proposed partnership is also a priority objective.

As a component of the Department's organic industrial base, DMEA is uniquely positioned to meet DoD demand by bringing online approximately 20,000 sq.ft of, newly constructed cleanroom space in a facility that can be leveraged to facilitate the proposed partnership. The

addition of SiC epitaxy capability will directly support DMEA's 150mm semiconductor fabrication facility in its mission to supply discrete high voltage components to DoD platforms.

DMEA envisions a flexible partnership agreement that provides maximum benefit to the commercial sector. For example, a proposal may stipulate that epi wafers produced in excess of DoD demand be sold by a prospective partner to commercial fabrication facilities. Other potential opportunities include technology (IP) licensing arrangements with external commercial entities, for technologies developed under the partnership. Contingent upon proposal specifics, it may be possible to expand the proposed partnership into ancillary areas such as leveraging DMEA's semiconductor manufacturing capacity for prototyping or lower-volume production. The objective is to develop a mutually beneficial agreement that emphasizes a tangible benefit to DoD while maintaining an equitable benefit for prospective commercial partners.

DMEA requests that interested respondents provide a white paper addressing the following topics

- Qualification- Describe in detail corporate experience and knowledge relevant to the growth of epitaxial films on 4H SiC substrates.
 - Include R&D efforts.
 - Include material that demonstrates awareness of the technical issues associated with the growth of thick epi layers.
 - Describe manufacturing capacity, if applicable.
 - Provide biographic/background information of key individuals involved in technology development, R&D, or process development relevant to SiC epitaxy.
 - Index of accepted articles, industry white papers, presentations relevant to SiC epitaxial growth, published in peer reviewed journals.
- Partnership Model- Describe in detail your organization's vision of a structured strategic partnership
 - Anticipated roles
 - Identify the anticipated benefits derived, by both parties, resultant from participation.
 - Identify the expected contributions from the prospective industry partner and what contributions would the proposing partner expect from the U.S Government.
 - Capital/Equipment
 - Labor
 - Facility/manufacturing capacity access
 - Skills/Knowledge
 - Other relevant experience
 - Partnership Management
 - Joint planning/road mapping
 - Communication strategy
 - Proposed milestone schedule
- Business Case-Clearly articulate how this partnership will achieve success and mitigate identified risk factors.

- Market Analysis-Demonstrate understanding of the economics of the global semiconductor industry and the position of SiC manufacturing in the ecosystem.
 - Quantify the projected size of the market, and the market share that will be serviced under the proposed partnership.
- Financial Analysis-Describe how the anticipated benefits outweigh the projected costs of engaging in the proposed partnership.
 - Describe how the proposed partnership will yield an acceptable return on investment for the commercial entity and provide tangible benefit to DMEA/DoD.
- Risk Management-Describe identified and potential risks for either partner engaging in the proposed partnership and potential mitigations.
 - Outline a strategy for proactive risk mitigation

3.0 TECHNICAL METRICS, OBJECTIVES, AND OUTCOMES: For proposal development purposes, the following substrate specifications will be used for starting material.

Parameters	Specification
Material	Silicon Carbide, (SiC)
Poly Type	4H
Diameter	150mm $\pm$ 0.2mm
Grade	Production Grade
Thickness	350 $\pm$ 25 μ m
Type/Dopant	N/Nitrogen
Resistivity	0.015 – 0.025 Ω -cm
Orientation, face	<11-20>OFF 4° $\pm$ 0.15°
Primary flat length	47.5 $\pm$ 1.5mm
Primary flat orientation	(1-100) $\pm$ 5°
Secondary flat	None
LTV	$\leq$ 3 μ m (5mm*5mm)
TTV	$\leq$ 5 μ m
Bow	-25~25 μ m
Warp	$\leq$ 35 μ m
Front finish	Si-face CMP
Back finish	C-face polished
Surface roughness (front and back)	RA $\leq$ 0.2nm (5 μ m*5 μ m)
Micropipe density	$\leq$ 0.2/cm ²
Metal impurities	$\leq$ 5E10 atoms/cm ²
BPD	$\leq$ 1500/cm ²
TSD	$\leq$ 300/cm ²
Particles (front side, size $\geq$ 0.3 μ m)	$\leq$ 60
Scratches (front)	$\leq$ 5/wafer, Total length $\leq$ Diameter
Scratches (back)	$\leq$ 5/mm, Total length $\leq$ 2*Diameter
Orange peel/pits/cracks/contamination/stains/striations (front side)	None
Edge chips/indent/fracture (front side)	None

Back side defects (edge chips/indents)	None
Polytype areas (front side)	None
Front laser marking	None
Back laser marking	1mm away from top edge
Edge	Chamfer

The initial technical objective of the proposed partnership is the manufacture of high quality SiC epitaxial wafers for high voltage and high temperature applications. The following general specifications describe the desired epitaxial growth capability to be developed under the proposed partnership.

Material	Thickness	Species	Doping (cm⁻³)	Wafer Useable Area (250μm)
Epi SiC	3μm-250μm	ntype, ptype	n=1E14-1E19, p=5E14-1E20	>65% (150mm wafer)

4.0 POINT OF CONTACT:

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